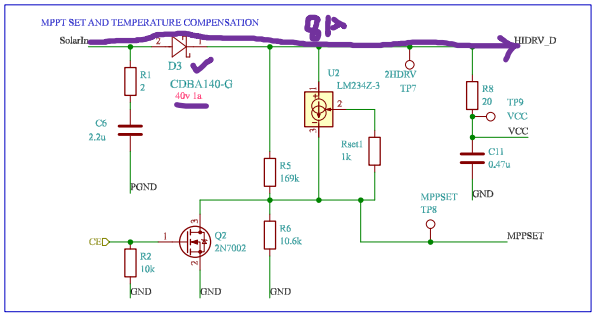
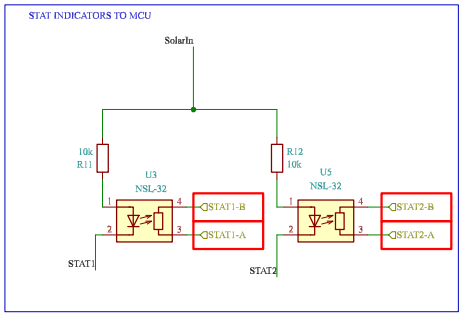
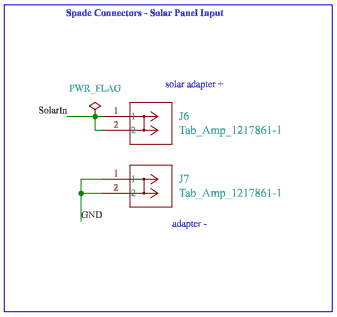
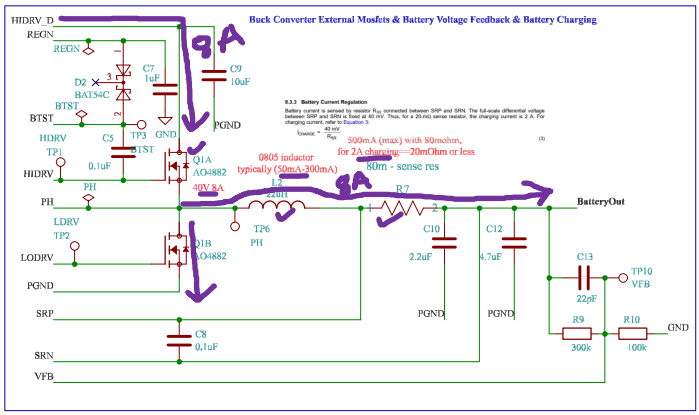
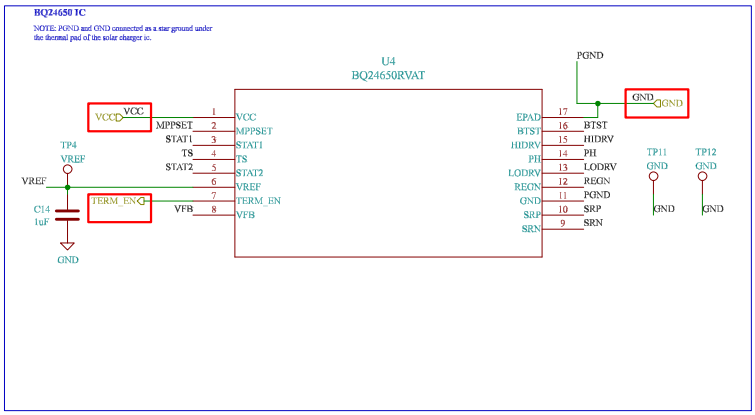
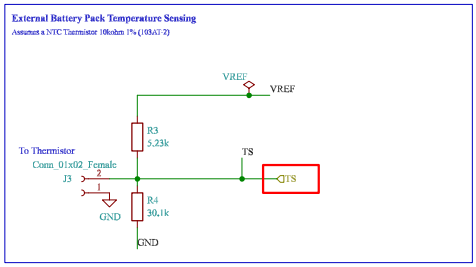

MPPT SET AND TEMPERATURE COMPENSATION
SolarIn
HIDRV_D
D3
CDBA140-G
40v 1a
R1
2
C6
2.2u
PGND
U2
LM234Z-3
2HDRV
TP7
R8
20
TP9
VCC
VCC
C11
0.47u
GND
Rset1
1k
R5
169k
MPPSET
TP8
MPPSET
CE
Q2
2N7002
R2
10k
R6
10.6k
GND
GND
GND

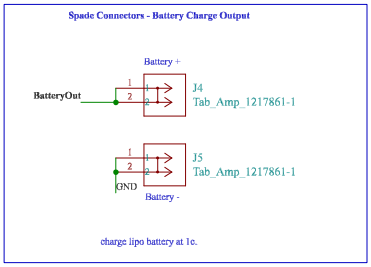

STAT INDICATORS TO MCU
SolarIn
10k
R11
R12
10k
U3
NSL-32
U5
NSL-32
STAT1-B
STAT1-A
STAT2-B
STAT2-A
STAT1
STAT2


Spade Connectors - Solar Panel Input
PWR_FLAG
solar adapter +
SolarIn
J6
Tab_Amp_1217861-1
J7
Tab_Amp_1217861-1
GND
adapter -


Buck Converter External Mosfets & Battery Voltage Feedback & Battery Charging
HIDRV_D
REGN
C7
1uF
C9
10uF
D2
BAT54C
BTST
PGND
C5
0.1uF
TP3
GND
HIDRV
TP1
Q1A
AO4882
40V 8A
0805 inductor
typically (50mA-300mA)
L2
22uH
500mA (max) with 80mohm,
for 2A charging—20mOhm or less
80m - sense res
R7
BatteryOut
PH
TP6
PH
LDRV
TP2
Q1B
AO4882
LODRV
PGND
C10
2.2uF
C12
4.7uF
PGND
PGND
C13
22pF
TP10
VFB
SRP
C8
0.1uF
R9
300k
R10
100k
GND
SRN
VFB


BQ24650 IC
NOTE: PGND and GND connected as a star ground under the thermal pad of the solar charger ic.
U4
BQ24650RVAT
PGND
VCC
MPPSET
STAT1
TS
STAT2
VREF
TERM_EN
VFB
EPAD
BTST
HIDRV
PH
LODRV
REGN
PGND
SRP
SRN
GND
TP4
VREF
C14
1uF
TP11
GND
TP12
GND


External Battery Pack Temperature Sensing
Assumes a NTC Thermistor 10kohm 1% (103AT-2)
VREF
R3
5.23k
To Thermistor
Conn_01x02_Female
J3
TS
R4
30.1k
GND
GND


Spade Connectors - Battery Charge Output
Battery +
BatteryOut
J4
Tab_Amp_1217861-1
J5
Tab_Amp_1217861-1
GND
Battery -
charge lipo battery at 1c.